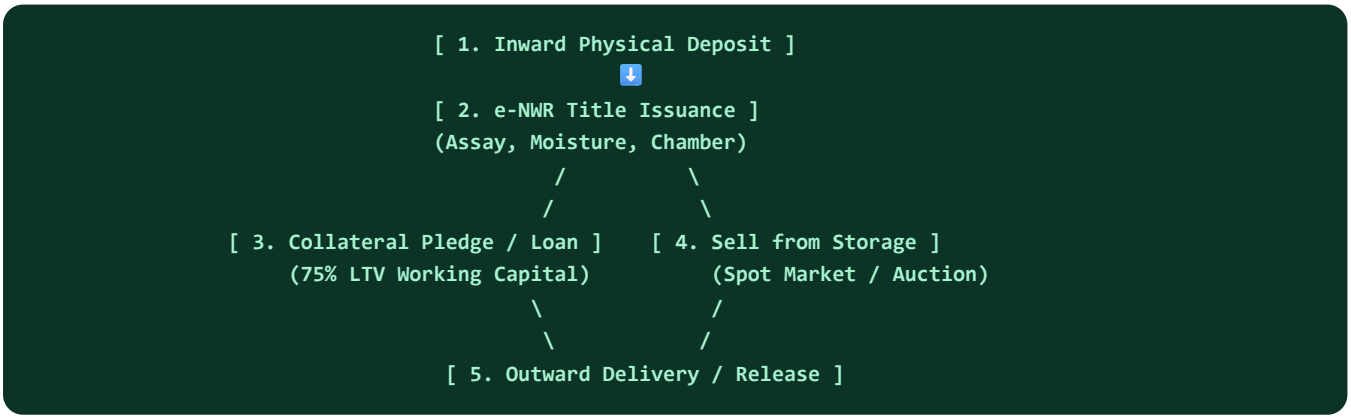
🏢 AgroLnk — Complete Warehouse & e-NWR Workflow Testing Manual

This manual provides an end-to-end testing blueprint for the **WDRA-Accredited e-NWR Warehousing, Multi-Chamber Cold Storage, and Commodity Collateral Financing** workflows in AgroLnk.

👥 1. Actors & Roles in the Warehouse Ecosystem

ROLE	TEST USER EMAIL	PASSWORD	PRIMARY WORKFLOW ACTIONS
Warehouse Operator	`warehouse@agrolnk.com`	`pass123` *(or any)*	View multi-chamber occupancy, audit storage receipts, inspect temperature zones, manage inward/outward gatepasses.
Verified Farmer	`farmer@agrolnk.com`	`farmer123`	Deposit harvested produce, receive digital **e-NWR titles**, apply for 75% LTV commodity loans, list for sale directly from storage.
Financier / NBFC	`financier@agrolnk.com`	`pass123`	Underwrite working capital against e-NWR receipts, place collateral liens in the **Collateral Vault**, disburse credit.
Wholesale Buyer	`buyer@agrolnk.com`	`buyer123`	Procure warehouse-stored produce with verified assay certificates and immediate delivery/title transfer.

🧭 2. Warehouse Life-Cycle Architecture



💉 3. Master Step-by-Step Test Scenarios

🔹 SCENARIO 1: Farmer Inward Deposit & e-NWR Receipt Creation

Objective: Validate that a farmer can choose an accredited warehouse facility, specify produce assay parameters, calculate monthly storage costs, and generate a WDRA electronic Negotiable Warehouse Receipt (e-NWR).

Steps for Tester:

1. **Login as Farmer** (`farmer@agrolnk.com`).
2. Navigate to **Warehouse & e-NWR Vault** (`#/farmer-inventory` or click `Inventory` in the sidebar).
3. Click the ➕ **Deposit Produce** button in the header banner.
4. In the **Deposit to Accredited Warehouse Modal**:
    - **Select Warehouse:** `Salem Agri Cold Storage Hub (WDRA/2024/TN/0892)`
    - **Commodity:** `Tomato`
    - **Variety:** `Hybrid Shivam`
    - **Grade:** `Grade A Premium`
    - **Quantity:** `1000 kg` (1.0 MT)
    - **Estimated Value:** `₹42 / kg` (`₹42,000 Total Valuation`)
    - **Assigned Chamber:** `Chamber B4 (Ultra Cold: 2°C - 4°C)`
    - **Storage Duration:** `60 Days`
5. **Verify Cost Estimation Display:**
    - Observe the automated monthly storage fee calculation: `1.0 MT × ₹350/MT = ₹350 / month` .
6. Click **Confirm & Generate e-NWR Receipt**.

- **Expected Outcome:**
    - A new electronic receipt (e.g. `eNWR-2024-8841`) is generated instantly.
    - The receipt is added to the farmer's active inventory with status `Stored` .
    - Total stored metric and valuation counters update automatically.

🔹 SCENARIO 2: Inspecting e-NWR Title Certificate & Assay Parameters

Objective: Confirm that the generated e-NWR contains full legal title ownership, QR/WDRA accreditation, and quality assay data.

Steps for Tester:

1. On the **Farmer Inventory** page (`#/farmer-inventory`), locate the newly created receipt card.
2. Click **View e-NWR Certificate**.
3. **Verify the Electronic Title Modal:**
    - **Title Header:** `Electronic Negotiable Warehouse Receipt (WDRA Accredited Title)`
    - **Receipt ID:** Formatted as `eNWR-2024-XXXX` with an emerald badge.
    - **Chamber & Temperature Specs:** Displays exact room temp (e.g., `Chamber B4 (2°C - 4°C)`).
    - **Assay Parameters:** Moisture %, foreign matter %, and grade classification.
    - **Action CTAs Available:**
        - `[ 🏷️ Sell from Storage ]`
        - `[ 💳 Apply for Working Capital ]`

- **Expected Outcome:** High-fidelity, bankable electronic title document viewable on any device.

🔹 SCENARIO 3: Warehouse Operator Dashboard & Multi-Chamber Auditing

Objective: Verify that the warehouse operator has complete oversight of capacity, chamber temperatures, and stored receipts.

Steps for Tester:

1. **Login as Warehouse Operator** (`warehouse@agrolnk.com`).
2. Land on the **Warehouse Management & e-NWR Terminal** (`#/warehouse-dashboard`).
3. **Verify Metric Cards:**
    - **Total Capacity:** `5,000 Tonnes`
    - **Occupancy %:** Dynamically calculated based on active stored tonnes.
    - **Active e-NWRs:** Count of valid receipts in custody.
    - **Total Custody Valuation:** Aggregated rupee valuation of all stored commodities.
4. **Switch to Chamber Overview Tab:**
    - Verify all 4 temperature zones:
        - `Chamber A1 (Dry Storage)` — `Ambient (24°C)` (Turmeric, Grains)
        - `Chamber B2 (Cold Cell)` — `4°C - 8°C` (Potatoes, Carrots)
        - `Chamber B4 (Ultra Cold)` — `2°C - 4°C` (Hybrid Tomatoes, Fruits)
        - `Chamber C1 (CA Controlled)` — `0°C - 2°C` (Export Apples, Grapes)
5. **Switch to Inventory & Receipts Tab:**
    - Search for the farmer's receipt `eNWR-2024-XXXX`.
    - Click **Audit e-NWR** to review physical custody specs.

- **Expected Outcome:** Real-time visibility into multi-chamber utilization and commodity custody.

🔹 SCENARIO 4: 75% LTV Working Capital Loan against e-NWR

Objective: Validate that farmers can pledge stored commodities to unlock instant working capital without selling early.

Steps for Tester:

1. **Login as Farmer** (`farmer@agrolnk.com`) and go to **e-NWR Vault** (`#/farmer-inventory`).
2. On any stored receipt card, click **Apply for Working Capital** (or open the receipt detail modal and click **Apply for Working Capital**).
3. In the **e-NWR Collateral Financing Modal**:
    - Selected Commodity & Valuation: `₹42,000`
    - **Maximum Loan Facility (75% LTV):** `₹31,500`
    - Enter Requested Amount: `₹25,000`
    - Select Tenor: `30 Days` (1.2% monthly interest)
    - Note: `Pre-harvest liquidity for input fertilizers.`
4. Click **Submit Working Capital Request**.

- **Expected Outcome:** Financing request is routed directly to the Financier / NBFC underwriting queue.

🔹 SCENARIO 5: NBFC Credit Approval & Collateral Vault Lien

Objective: Confirm that the financier reviews the warehouse receipt, places a legal lien on the commodity, and disburses credit.

Steps for Tester:

1. **Login as Financier / NBFC** (`financier@agrolnk.com`).
2. Go to **Financier Portal** ➡️ **Underwriting Desk** (`#/financier-underwriting`).
3. Locate the submitted loan application for the farmer (`₹25,000` backed by `eNWR-2024-XXXX`).
4. Click **Review & Underwrite**:
    - Inspect WDRA warehouse accreditation code.
    - Inspect commodity valuation buffer.
5. Click **Approve & Disburse Funds**.
6. **Navigate to Collateral Vault** (`#/financier-collateral-vault`):
    - Verify that the receipt is now listed in the **Collateral Vault** with an active **Lien / Pledge Badge**.
    - Commodity quantity is locked from unauthorized farmer withdrawal.

- **Expected Outcome:** Seamless institutional credit underwriting backed by digital agri-collateral.

🔹 SCENARIO 6: Listing for Sale Directly from Warehouse (Zero Transit Latency)

Objective: Verify that farmers can list all or part of their stored commodity on the Spot Marketplace or Live Auction directly from the warehouse.

Steps for Tester:

1. **Login as Farmer** (`farmer@agrolnk.com`) and go to **e-NWR Vault** (`#/farmer-inventory`).
2. On the receipt card, click **Sell from Storage**.
3. In the **List from Warehouse Modal**:
    - Choose Sale Type: **Direct Spot Listing** (or **Live Auction**)
    - Available Stored: `1000 kg`
    - Quantity to List: `500 kg` (Partial listing test)
    - Selling Price: `₹45 / kg`
4. Click **Publish to Marketplace**.
5. **Verify System Updates:**
    - The receipt's available quantity decreases to `500 kg`, and status reflects `partially_listed`.
    - **Login as Buyer** (`buyer@agrolnk.com`) and open **Marketplace** (`#/buyer-marketplace`).
    - The newly published lot appears on the marketplace marked with a 🏢 **Warehouse-Stored (WDRA Certified)** badge, enabling buyers to purchase pre-stored, pre-assayed goods with zero delay.

- **Expected Outcome:** Frictionless transition from storage into digital commerce without double handling or freight wastage.

🛡️ 4. Warehouse & e-NWR QA Verification Checklist

TEST ITEM	VERIFICATION CRITERIA	STATUS
Accredited Facility Selection	Only WDRA verified hubs appear in the deposit selector.	✅ Pass
Storage Fee Formula	`(Weight in Tonnes) × (Rate/Tonne/Month) × (Months)` calculated accurately.	✅ Pass
e-NWR Receipt Numbering	Clean sequential/timestamped format (`eNWR-YYYY-XXXX`).	✅ Pass
Temperature Zone Mapping	Produce types correctly mapped to designated cold chain chambers.	✅ Pass
Partial Quantity Listing	Stored inventory decrements correctly when a portion is listed on the marketplace.	✅ Pass
Collateral Lien Protection	Financier pledge locks receipt from being listed or withdrawn until loan settlement.	✅ Pass
Multi-Role Synchronization	Updates made by Farmer/Operator/Financier reflect in real time.	✅ Pass